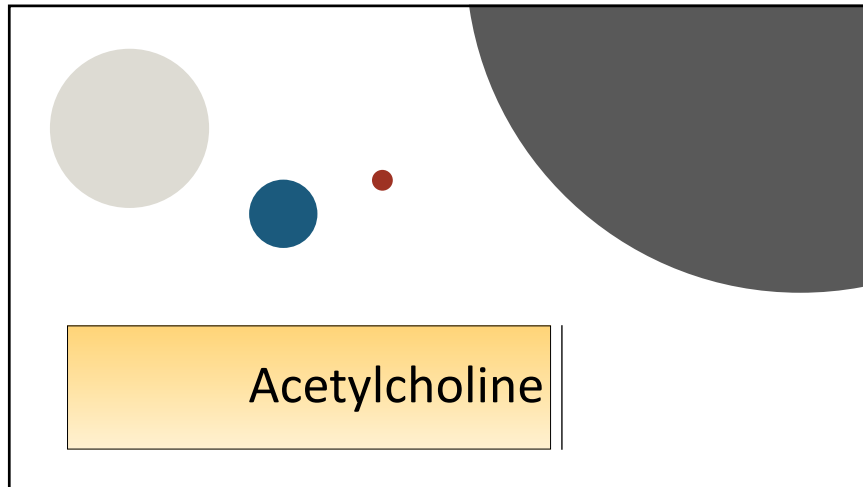
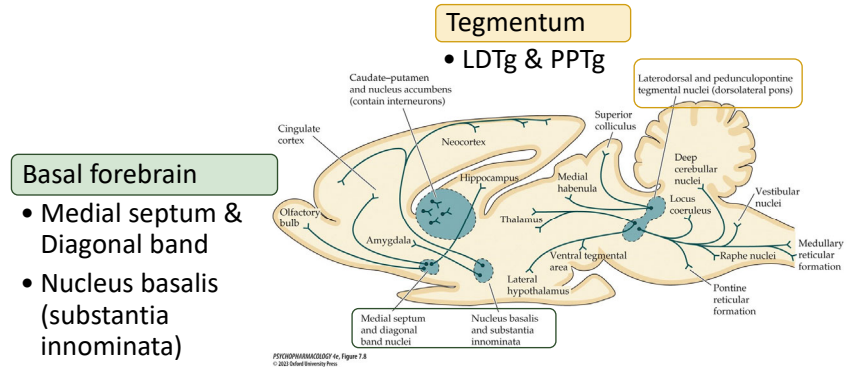
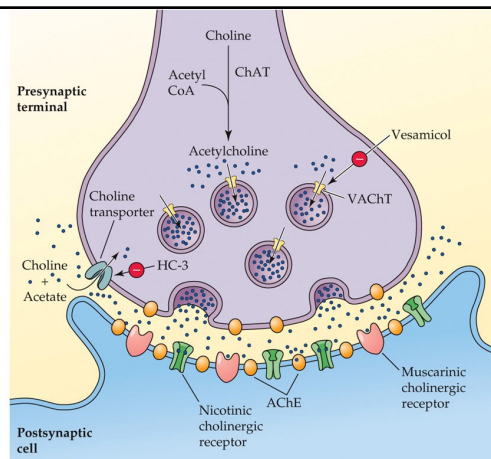




Cholinergic neurons in the CNS



Synthesis, Release & Inactivation



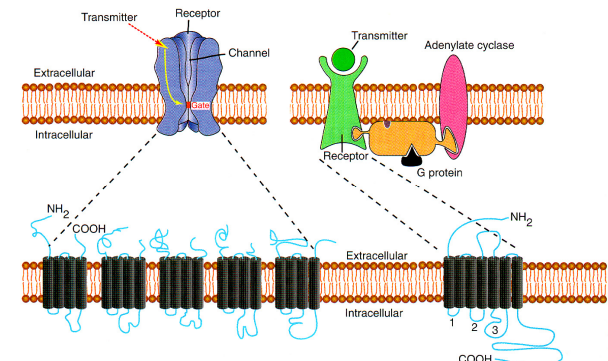
ACh activates ionotropic & metabotropic R's

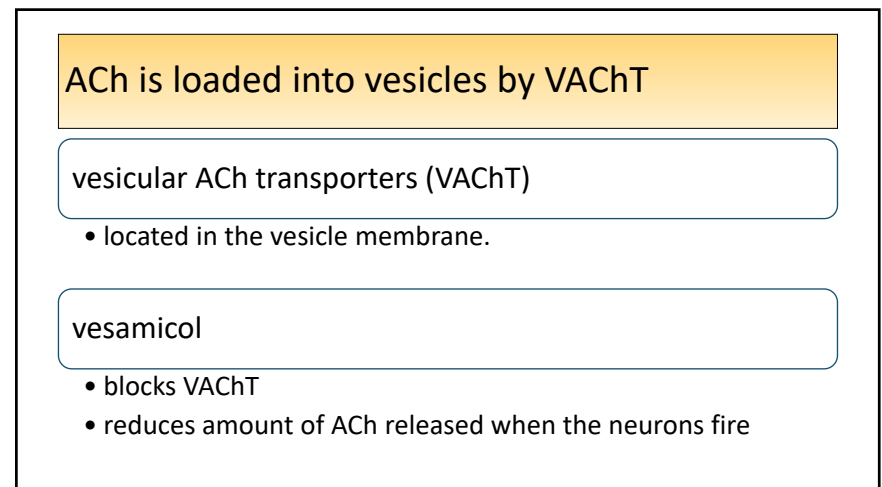
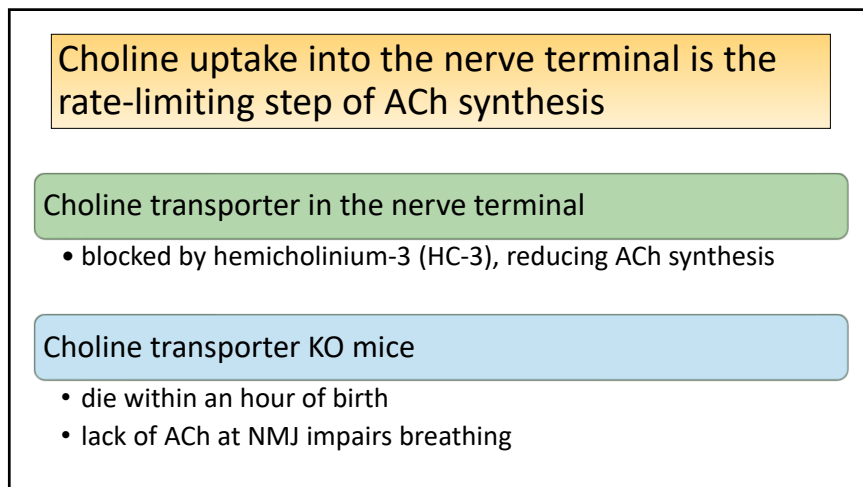
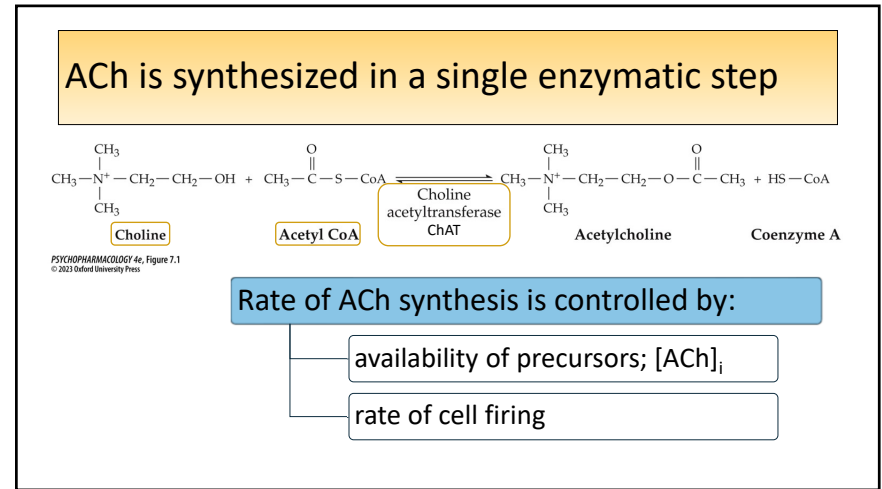
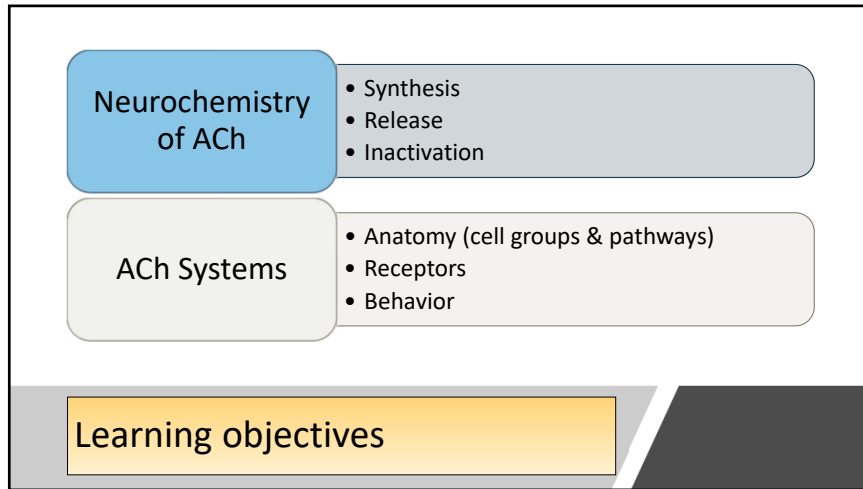
Ionotropic

- nicotine
- curare

Metabotropic

- muscarine
- atropine





ACh release is affected by various toxins

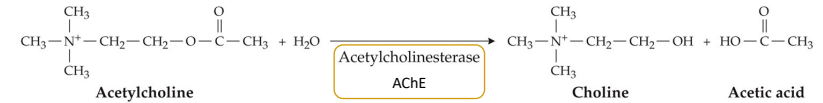
Black widow spider venom

- massive release of ACh in the PNS.
- muscle pain, tremors, nausea, vomiting, salivation, copious sweating

Botulinum toxin

- Clostridium botulinum*
- inhibits ACh release
- BoTox - local paralysis to reduces wrinkles

ACh is degraded by AChE



Acetylcholinesterase (AChE) is located

- in the presynaptic neuron
- on the postsynaptic membrane
- in the basal lamina of NMJ

AChE is the target of some beneficial substances and some deadly substances

Alzheimer's drugs

- reversible AChE inhibitors
- donepezil
- rivastigmine

Other clinically useful substances

- Physostigmine (crosses the blood-brain barrier). Used to treat glaucoma.
- Neostigmine and pyridostigmine (do not cross the blood-brain barrier). Used to treat myasthenia gravis, an autoimmune disorder.

AChE is the target of some beneficial substances and some deadly substances

Organophosphates

- irreversible AChE inhibitors
- insecticides
- nerve agents (soman, sarin, VX)

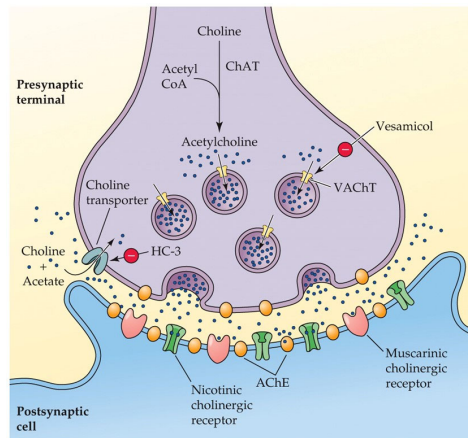
Exposure results in

- profuse sweating
- salivation
- vomiting
- loss of bladder and bowel
- convulsions
- death by asphyxiation

Chapter 07 – Acetylcholine

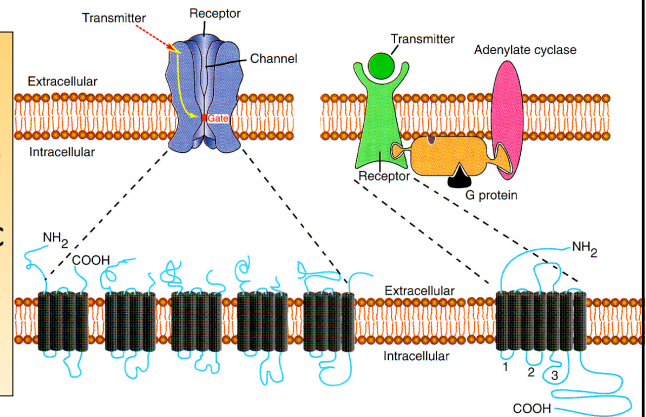
Summary of Acetylcholine

Synthesis, Release & Inactivation



PSYCHOPHARMACOLOGY 4e, Figure 7.2
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ACh activates ionotropic & metabotropic receptors



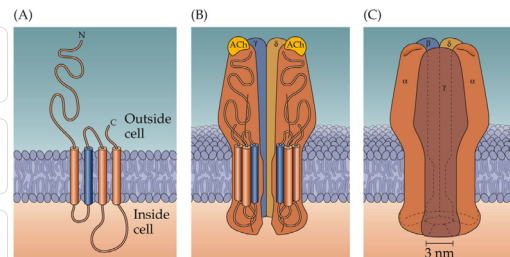
ACh activates ionotropic & metabotropic receptors

nAChR

5 subunits arranged around a central pore

Each subunit has 4 TMD's

TMD 2 (M2) lines the pore

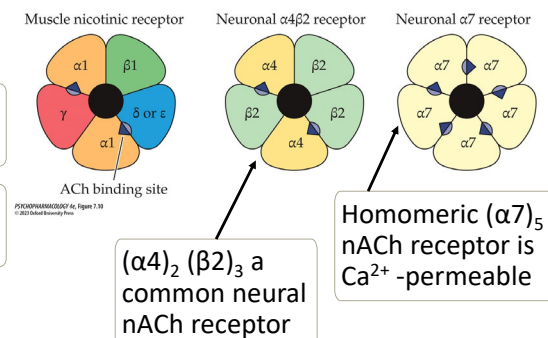


ACh activates ionotropic & metabotropic receptors

nAChR

Agonists: **nicotine** & **succinylcholine**

Antagonists: **curare** & **mecamylamine**



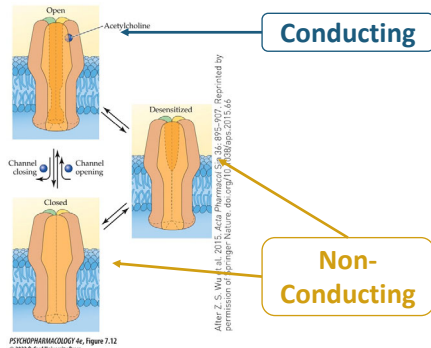
ACh activates ionotropic & metabotropic receptors

3 transition states

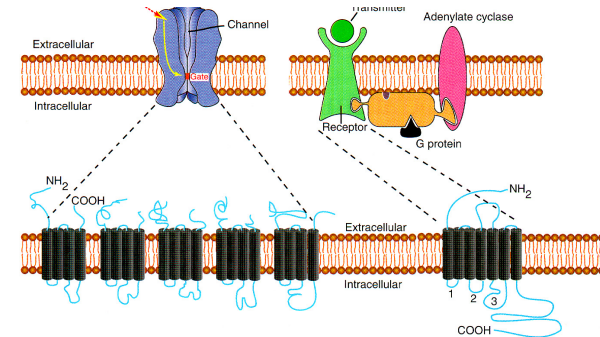
closed

open

desensitized



ACh activates ionotropic & metabotropic receptors



ACh activates ionotropic & metabotropic receptors

mAChR

7 TMD, GPCR

Agonists: **muscarine** & **pilocarpine**

Antagonists: **atropine** & **scopolamine**

5 subtypes

M1 – M5

LDTg input to VTA activates M5, increases DA neuron firing, behavioral relevance of stimuli

Parasympathetic control of cardiac muscle involves M2

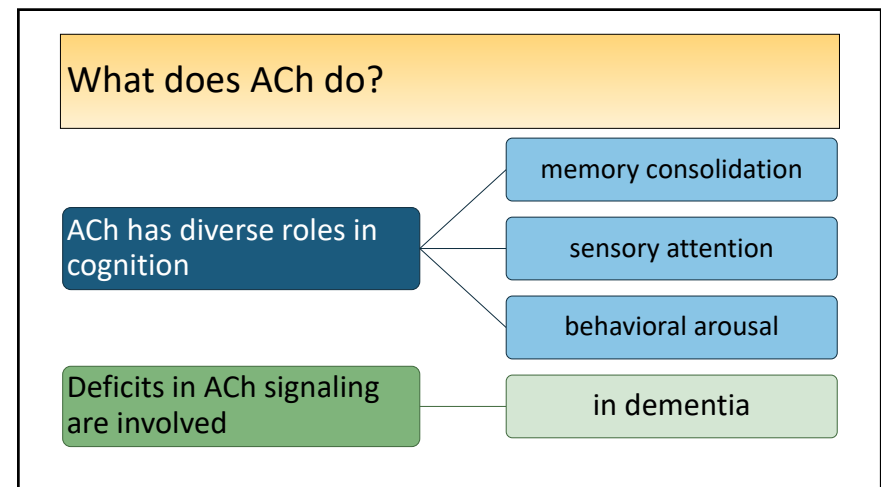
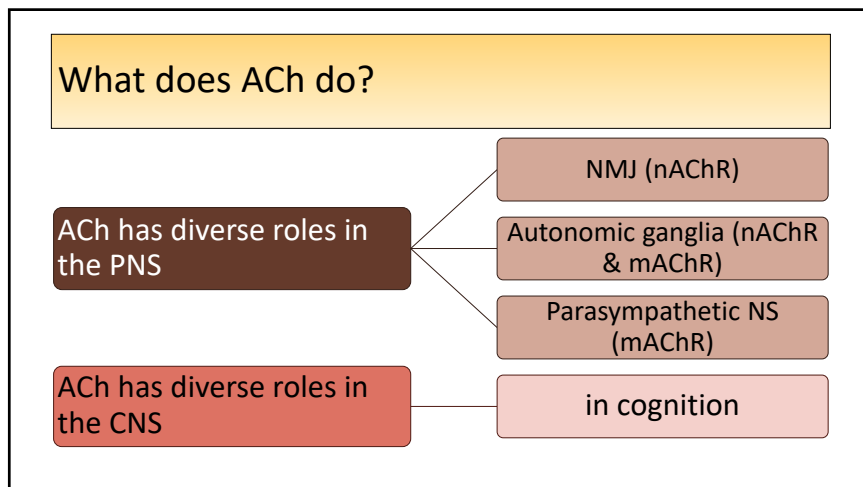
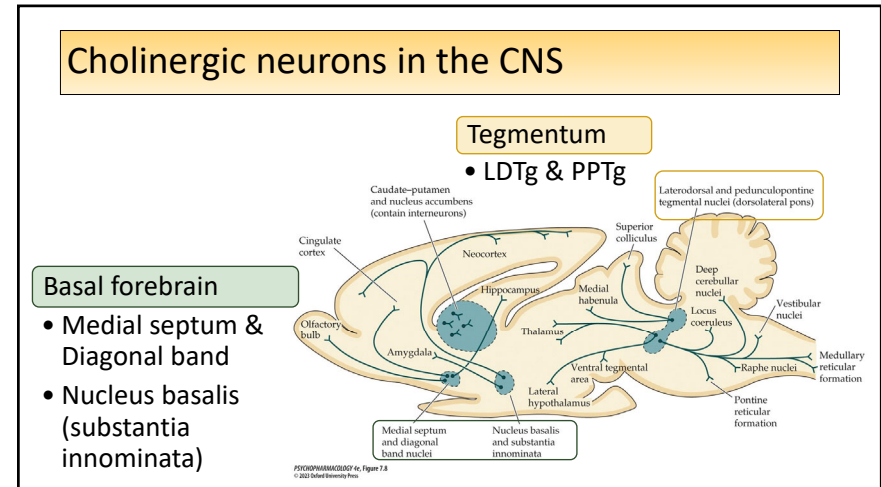
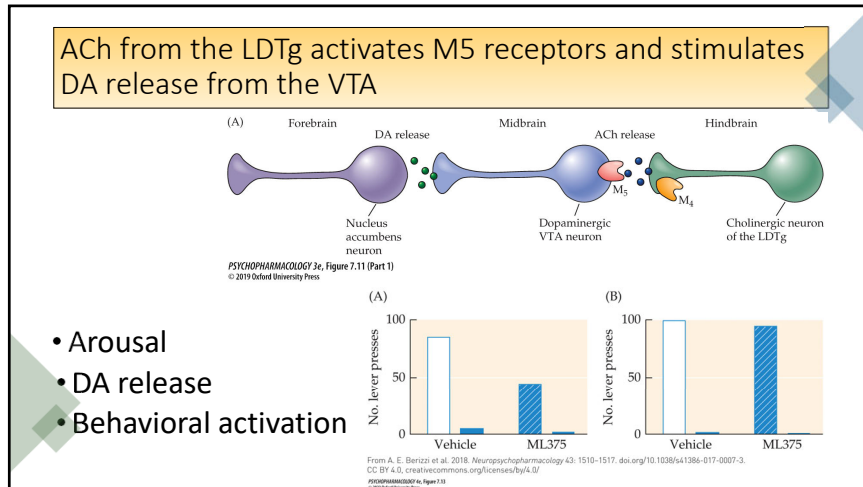
Summary of Acetylcholine Neurotransmission

Synthesis, storage and release, inactivation

- Choline; ChAT; VACHT; AChE

AChE is used at NMJ, in PNS & in CNS

- nAChR (ionotropic) and mAChR (metabotropic)
- All preganglionic ANS; postganglionic parasympathetic
- Modulates firing of DA neurons in VTA, cortical & behavioral arousal



mACh receptor antagonists are amnesic

Scopolamine or atropine impair performance on learning tasks

- classical conditioning, discrimination learning, spatial learning
- active and passive avoidance tasks, delay tasks

mACh receptor antagonists have been used to model memory deficits in AD

- exacerbate cognitive impairments of patients with AD

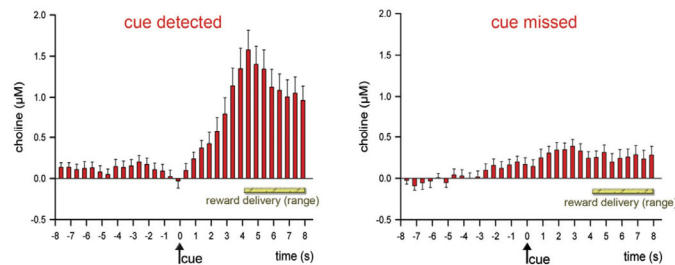
Memory impairments may result from deficits in sensory attention

ACh is released in PFC

- from the basal forebrain cholinergic system
- phasic (or transient) release
- associated with detection of cues (“hits”) in attention tasks



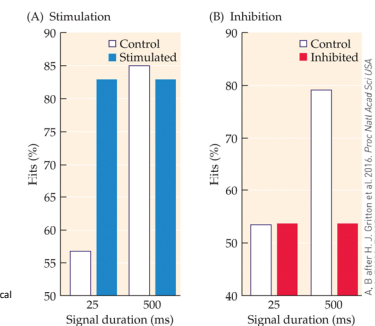
ACh “transients” in PFC signal detection of a cue



Parikh, V., Kozak, R., Martinez, V. & Sarter, M. (2007). Prefrontal acetylcholine release controls cue detection on multiple timescales. *Neuron*, 56, 141-154. doi: 10.1016/j.neuron.2007.08.025

Optogenetic stimulation of BFCS improves “hit rate,” inhibition of BFCS impairs “hit rate.”

Phasic cholinergic input to the PFC is used to guide behavioral choice in response to external cues



Gritton, H. J., Howe, W. M., Mallory, C. S., Hettrich, V. L., Berke, J. D. & Sarter, M. (2016). Cortical cholinergic signaling controls detection of cues. *Proceedings of the National Academy of Sciences of the United States of America*, 113, E1089-E1097. doi: 10.1073/pnas.1516134113

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Summary: Behavioral pharmacology of ACh

The *attentional hypothesis*

- Arousal
 - cortical potentiation
 - Subcortical – sleep; behavioral activation
- Attention to stimuli guiding behavioral responses
- Memory